

Madi (Madeline) Abio

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EDUCATION

Griffith University <i>Bachelor's of Computer Science / Bachelor's of Electrical Engineering (Honours)</i> • Cumulative GPA: 6.32/7	Gold Coast, Queensland Mar 2022 — Nov 2026
University of Utah <i>International Trimester Exchange</i>	Salt Lake City, Utah, USA 2024

AWARDS & SCHOLARSHIPS

- **Academic Excellence** | Griffith University, 2025
- **Academic Excellence** | Griffith University, 2024
- **Academic Excellence** | Griffith University, 2023
- **Chancellor's Scholarship** | Griffith University, 2025
- **Amplify Connector** | Energy Queensland, 2025 – Present
- **Future of Energy Scholarship** | Energy Queensland, 2023 – 2025
- **Students in Power Bursary** | Australian Institute of Power, 2022 – 2023
- **Brighter Futures Scholarship** | The Abedian Foundation, 2022

WORK EXPERIENCE

Software Engineer VPTech	Aug 2025 — Present Gold Coast, QLD
• Developing production-grade agentic Retrieval-Augmented Generation (RAG) systems with human-in-the-loop feedback and validation, including Vector Database integration, MCP tool development and interaction/interface design (AWS AgentCore, LangGraph, LangChain). • Designing and implementing core full-stack features using React and NestJS, contributing to system design, structural conventions and tooling decisions. • Improved development velocity and quality by improving CI/CD pipelines and introducing automated E2E/unit testing (Playwright, Vitest) and code quality (QA) checks (ESLint, CDK-Nag). • Oversaw the integration of Jira and Agile project management strategy, optimising for rapid development.	
Grid Technology Intern Energy Queensland	Jan 2025 — Sept 2025 Brisbane, QLD
• Developed an anomaly detection machine learning model to detect neutral faults targeting deployment across 10,000 smart meters (see: Neutral Fault Detection project). • Streamlined daily operational data analysis by building a solar panel data dashboard in Python/Power BI, reducing manual processing time from 15 minutes to 15 seconds. • Managed project delivery using Agile principles, including facilitating weekly sprints. • Automated various non-technical workflows. • Coordinated two internal networking events for interns and graduates.	
Undergraduate Research Assistant Griffith University	Mar 2024 — Jun 2024 Gold Coast, QLD
• Performed data pre-processing for a machine learning research project by cleaning and structuring large datasets using Pandas (Python), Bash and the HDF5 format.	
Cyber Security Platforms Intern Energy Queensland	Nov 2023 — Feb 2024 Brisbane, QLD
• Developed a strong foundation in computer networking through self-directed training and mentorship, covering the OSI model, network protocols, and enterprise network architecture. • Resolved ServiceNow tickets by applying network and security protocol knowledge. • Gained exposure to enterprise cybersecurity practices, including cloud security, firewall management (Panorama), and Netskope.	

PROJECTS

Full-Stack LeetCode & Codeforces Web App (madiab.io)

TypeScript, March 2026

- Designed a full-stack web app to host my LeetCode solve times. Extended this to also host my portfolio.
- NestJS backend, Next.js frontend, PostgreSQL database, Prisma ORM and Better Auth for authentication.
- Dynamically linked the projects section to my GitHub repositories via their public API.
- Hosted on Railway.

Digital Bass Synthesizer on Microcontroller

Embedded C, Sept 2025 – Oct 2025

- Designed a digital bass synthesizer running at a 48 kHz sample rate on the TM4C1294NCPDT microcontroller.
- Developed firmware in bare-metal embedded C to generate waveforms, control ADSR envelopes, and handle user input from potentiometers (via analog-to-digital converters) and matrix keypad.
- Engineered a custom external clock circuit to bridge the I²C peripheral for compatibility with the I²S DAC.
- Configured SPI peripheral to drive the 240x320 LCD for waveform and real-time ADSR value visualization.
- Implemented DMA ping-pong buffers for continuous audio streaming and optimized ISR timing for stable waveform output.

Neutral Fault Detection (Anomaly Detection)

Python, Feb 2025 – Sept 2025

- Developed an in-house neutral fault detection algorithm based on an internal research paper to investigate excessive false positives produced by a third-party solution that also lacked coverage across all smart meters.
- During implementation and side-by-side evaluation, identified systematic false positives in the third-party system on solar PV sites, caused by violated voltage–current modelling assumptions that made normal solar behaviour resemble neutral fault signatures.
- Proposed a revised approach that explicitly accounted for solar PV electrical behaviour.
- Designed and prototyped a scalable architecture targeting deployment across 10,000+ smart meters.
- Built a parallelized feature extraction and model evaluation pipeline, and applied time-ordered, meter-isolated validation with systematic error and failure-mode analysis.

Encryption Client

C++, May 2025

- Developed a terminal-based multi-round message encryption/decryption client, applying OOP principles with abstract base classes, inheritance, and polymorphism for modular, reusable design.
- Implemented a finite-state machine menu interface with input validation and exception handling, leveraging STL and lambda functions for efficient message processing.

CLUBS & SOCIETIES

- **Ravi's Study Program** | *Mentee, 2026*
- **Griffith University Coding Club (GCC)** | *ICPC Div B Competitor & Member, 2025 – Present*
- **UQ Computing Society** | *Lead Guitar Player (UQCS Band), Competitive Programmer, 2025 – Present*
- **UQ Fintech Society** | *Member, 2025 – Present*
- **Griffith University Advanced Robotics Development (GUARD)** | *Member, 2023 – Present*
- **Griffith University Gold Coast IEEE Student Branch** | *Administrator, 2023*

SKILLS

- **Programming Languages:** Python, TypeScript, C, C++, C#, Bash
- **Software Development & Tools:** Linux, React, Next.js, NestJS, PostgreSQL, Prisma, AWS, CI/CD, Git
- **Machine Learning & Data Science:** Applied Machine Learning, LLM orchestration, Data Science, AgentCore, Pandas, NumPy, scikit-learn, h5py, LangGraph, LangChain
- **Embedded & FPGA:** Embedded/Bare Metal C, SystemVerilog, PCB Design
- **Software Practices:** Object-Oriented Programming, Test-Driven Development
- **Professional Skills:** Technical Writing, Problem Solving, Teamwork, Communication